

# ABDUL RAFAY

Karachi, Pakistan

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## PROFESSIONAL SUMMARY

Results-driven Deep Learning Engineer and Artificial Intelligence graduate focused on architecting and deploying production-grade computer vision models. Keen to consistently master emerging technologies and excel in my respective fields by engineering robust identification, authentication, and multimodal text processing pipelines to solve complex enterprise challenges.

## TECHNICAL SKILLS

**Programming Languages:** Python

**Machine Learning:** Model Training, Fine Tuning, Prompt Engineering

**Deep Learning:** RNNs, CNNs, Transfer Learning, SmolVLM, LLaMA 3.1, Qwen2.5, U2-Net

**Computer Vision:** Image Classification, Multimodal OCR, Custom Segmentation, Document Parsing

**Natural Language Processing:** Text Classification, Transliteration Frameworks, Information Extraction

**Data Analysis & Frameworks:** NumPy, Pandas, Matplotlib, FastAPI, PyTorch, Hugging Face

## WORK EXPERIENCE

**Unikrew Solutions** – Karachi, Pakistan

**February 2026 – Present**

*Deep Learning Engineer*

- Responsible for the complete end-to-end R&D lifecycle; researching academic literature, training advanced neural architectures, and developing AI projects entirely from scratch to deploy robust systems directly into production environments.
- Collaborated with senior engineers to develop a hybrid Urdu-to-Roman transliteration system by fine-tuning LLaMA 3.1 8B on a custom-built parallel corpus and integrating a gazetteer for proper noun lookup; outperformed ChatGPT on accuracy, standardization, and consistency, enabling fully automated transliteration at scale.
- Designed and trained a novel Multiscale Attention Fusion Network for deepfake detection combining frequency-domain, spatial, and patch-based branches; achieved 97.45% average accuracy across StyleGAN, Stable Diffusion, NanoBana, ChatGPT-generated, and real images, surpassing SOTA baselines; enabling automated identity authenticity verification and eliminating manual review.
- Consistently designing, testing, and optimizing bilingual English and Urdu OCR engines for high-fidelity document metadata extraction under real-world scanner variances.
- Leading the engineering deployment of highly secure identification and identity authentication projects to automate enterprise document integrity workflows.

**Unikrew Solutions** – Karachi, Pakistan

**July 2025 – January 2026**

*Machine Learning Intern*

- Built a CNIC verification pipeline that validates Pakistani identity documents against security features and detects photocopy or digitally tampered forgeries using the in-house deepfake model; eliminated manual CNIC verification and reduced fraud risk in onboarding workflows.
- Developed a document integrity detection system using U2-Net segmentation to identify watermarks, blurred regions, and manipulations in scanned bills and official documents; automated flagging and rejection of forged submissions, reducing manual document review overhead.
- Contributed to bug fixes, code refactoring, and performance improvements across multiple production AI pipelines in close collaboration with the engineering team.

## TECHNICAL PROJECTS

### OCR on National Identity Cards

*Technologies: SmolVLM (2B), PyTorch, Hugging Face, Multi-modal Computer Vision*

- Developed a multi-modal text extraction framework to structuralize and pull information from documents containing both complex English text and Urdu Nastaliq scripts.

- Fine-tuned the SmolVLM 2-Billion parameter vision-language model to accurately resolve script alignment and low-resolution artifact issues.

**Financial Bank Statement Parsing Engine**

*Technologies: Qwen2.5 (3B), Prompt Engineering, LLM Information Extraction, Python Data Ecosystem*

- Designed and built an end-to-end extraction pipeline to parse multi-page, unstructured physical and digital bank statements into clean, structured data profiles.
- Developed advanced prompt engineering techniques and deterministic constraints to guarantee absolute accuracy and consistent output structuring from the 3-Billion parameter Qwen backbone, eliminating model hallucination risks.

**EDUCATION**

|                                                                                          |                         |
|------------------------------------------------------------------------------------------|-------------------------|
| <b>Bahria University</b> – Karachi Campus                                                | <b>Graduation: 2026</b> |
| <i>Bachelor of Science in Artificial Intelligence (CGPA: 3.04 / 4.00)</i>                |                         |
| <b>Core Coursework:</b> Machine Learning, Deep Learning, Computer Vision Pipeline Design |                         |
| <b>Government College for Men</b> – Nazimabad                                            | <b>2021</b>             |
| <i>Intermediate (Pre-Engineering)</i>                                                    |                         |
| <b>The Smart School</b>                                                                  | <b>2019</b>             |
| <i>Matriculation (Science)</i>                                                           |                         |

**ADDITIONAL DETAILS**

**Soft Skills:** Strategic Sales Planning, Team Collaboration, Client Relationship Management, Adaptability, Problem Solving, Jira, Team Communication

**Languages:** English, Urdu, Punjabi

**Personal Background:** Born October 31, 2003, Nationality: Pakistani

**PROFESSIONAL REFERENCES**

**Muneer Afzal** – Senior Manager - International Risk, HBL  
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